

What is Cloud Computing?

Cloud computing is the **delivery of computing services over the internet** instead of using a local computer or physical servers.

These services include:

- Storage (saving files online)
- Servers (running applications)
- Databases
- Networking
- Software

In simple terms, cloud computing allows users to **store data and run applications from anywhere using the internet**.

Example: Using Google Drive to store files instead of saving them on your computer.

Types of Cloud Computing

1. **Public Cloud**
Services are available to everyone over the internet (e.g., Azure, AWS).
 2. **Private Cloud**
Used by a single organization for better security.
 3. **Hybrid Cloud**
Combination of public and private cloud.
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Key Characteristics

- **On-demand self-service** – Users can access resources whenever needed
 - **Scalability** – Resources can be increased or decreased easily
 - **Resource pooling** – Multiple users share the same resources
 - **Rapid elasticity** – Quick expansion of services
 - **Measured service** – Pay only for what you use
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What is Azure Cloud?

Microsoft Azure (Microsoft Azure) is a **cloud computing platform and service** created by Microsoft.

It helps developers and businesses to:

- Build and deploy applications
- Store and manage data
- Run virtual machines
- Use AI and analytics tools

Azure provides services through Microsoft data centers located across the world.

Azure Services (Examples)

- **Virtual Machines (VMs)** – Run operating systems in the cloud
 - **Azure Storage** – Store structured and unstructured data
 - **Azure SQL Database** – Cloud-based database service
 - **Azure App Services** – Build and host web apps
 - **Azure AI** – Machine learning and AI tools
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Advantages of Cloud Computing

- **Cost Efficient** – No need to buy expensive hardware
 - **Scalable** – Easily increase or decrease resources
 - **Accessible** – Access data from anywhere
 - **Backup & Recovery** – Data is safely stored and recoverable
 - **Automatic Updates** – No need for manual software updates
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Disadvantages of Cloud Computing

- **Requires Internet** – Cannot work without internet connection
 - **Security Risks** – Data breaches possible if not secured
 - **Limited Control** – Less control over infrastructure
 - **Downtime** – Service outages may occur
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5 Use Cases of Cloud Computing

1. Education

Platforms like Google Classroom use cloud to store notes, assignments, and conduct online classes.

2. E-Commerce

Online shopping websites use cloud to manage product data, payments, and customer information.

3. Healthcare

Hospitals store patient records in the cloud for easy and secure access.

4. Data Analytics

Companies use cloud computing to analyze large amounts of data and make business decisions.

5. Entertainment & Streaming

Services like Netflix use cloud to deliver movies and shows globally.